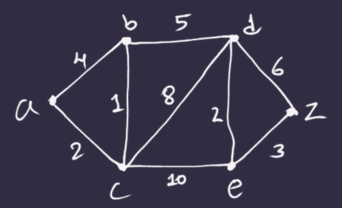
* Dijkstra's Algorithm

1. Assign source vertex to S.
   a) set $W(S) = 0$
   b) set $W(V) = \infty$ [for all vertices except S]
2. Suppose m be a vertices in graph for which $W(m)$ is minimum.
3. Make vertices m as visited.
4. Find adjacent vertices to I which are not visited.
5. Update weight of vertices of set I,

$$W(I_K) = \min\left(\underline{W(I_K)}, \underline{W(m) + W(m, I_K)}\right)$$

6. Repeat process from 3 to 5. until all vertices visited.

Q. Find shortest path from a to z
S



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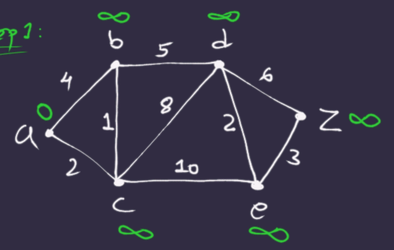
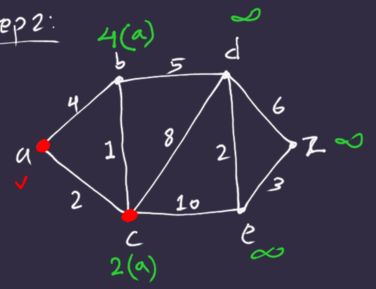
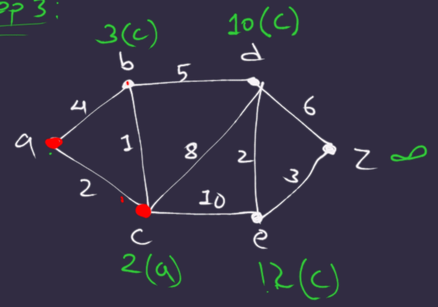
Step 1:



Step 2:



Step 3:



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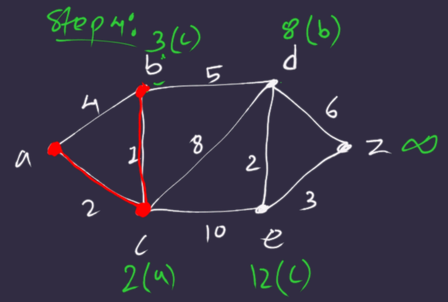

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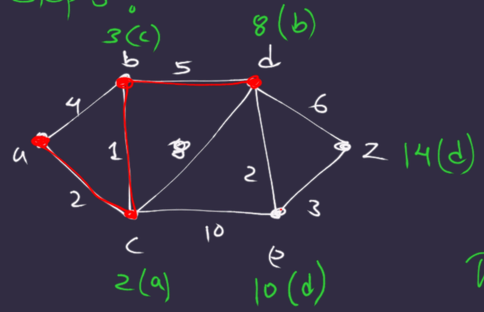

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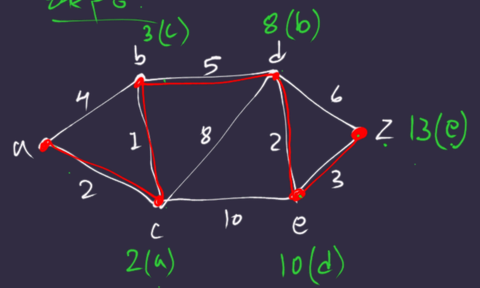
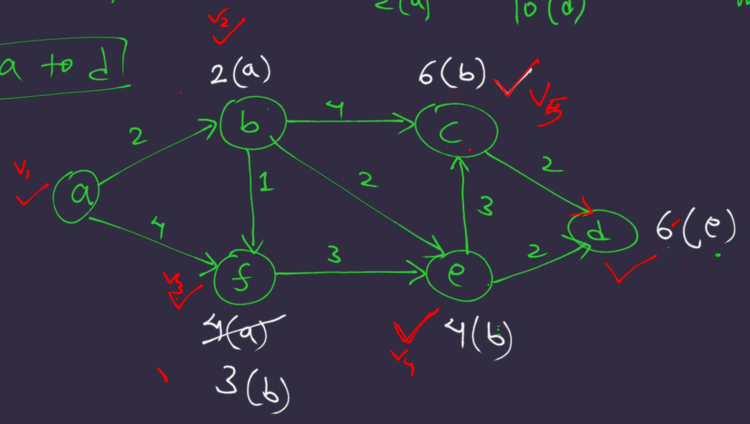


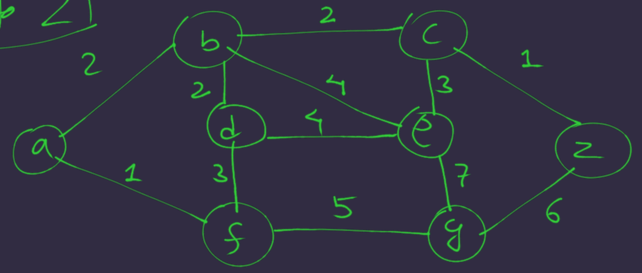
Now, shortest path from a to z is,

a – c – b – d – e – z

Total weight = 13

Q. $\boxed{a \text{ to } d}$



$a \xrightarrow{2} b \xrightarrow{2} e \xrightarrow{2} d$ 6(e)

Q. $\boxed{a \text{ to } z}$